

(1) Calculate the coefficient of determination R^2 as a function of the angle θ for a pure rotation. Use the data set at the right. Plot your results and determine the maximum and minimum values of the function. The pure rotation is described by:

$$\hat{x}_i = x_i \cos \theta - y_i \sin \theta \quad \text{and} \quad \hat{y}_i = x_i \sin \theta + y_i \cos \theta$$

where the circumflexes indicate the coordinate values in the rotated system.

(2) Derive the least squares method for a linear equation, *viz.*, $y = a x + b$ using horizontal offsets. Find expressions for the two coefficients in terms of sums defined as s_x , s_y , s_{xx} , *etc.* Fit the data at the right for both vertical offsets and horizontal offsets and compare the results. Why do you think they are different?

(3) Murphy 1.2 Solve using Cramer's rule.

(4) Murphy 1.15

(5) Murphy 1.17

x	y
1	4.5
2	11
3	8
4	13.9
5	12
6	14
7	18
8	25
9	19
10	23.6